Problem set 3

Exercise 1

Consider the following labelled training points in R^2:

Point	(x, y)	Class
A	$(1, 1)$	Red
B	$(2, 1)$	Red
C	$(4, 3)$	Blue
D	$(5, 4)$	Blue

Using the k**-Nearest Neighbours** classifier with $k = 3$ and **Euclidean distance**, determine the predicted class of the test point

$$Q = (3, 2).$$

Show your distance calculations and the majority vote among the k nearest neighbours.

Solution.

We use the Euclidean distance:

$$d\big((x_1, y_1), (x_2, y_2)\big) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}.$$

The test point is

$$Q = (3, 2).$$

Step 1: Compute distances.

$$d(Q, A) = \sqrt{(3-1)^2 + (2-1)^2} = \sqrt{4+1} = \sqrt{5} \approx 2.24,$$

$$d(Q, B) = \sqrt{(3-2)^2 + (2-1)^2} = \sqrt{1+1} = \sqrt{2} \approx 1.41,$$

$$d(Q, C) = \sqrt{(3-4)^2 + (2-3)^2} = \sqrt{1+1} = \sqrt{2} \approx 1.41,$$

$$d(Q, D) = \sqrt{(3-5)^2 + (2-4)^2} = \sqrt{4+4} = \sqrt{8} \approx 2.83.$$

Step 2: Sort by distance.

Point	Distance	Class
B	1.41	Red
C	1.41	Blue
A	2.24	Red
D	2.83	Blue

Step 3: Select the $k = 3$ nearest neighbours.
Nearest three points: B, C, A.

$$\text{Classes} = \{\text{Red}, \text{Blue}, \text{Red}\}$$

Step 4: Majority vote.

$$\text{Red} = 2, \quad \text{Blue} = 1$$

$$\boxed{\text{Predicted class of } Q \text{ is Red.}}$$